MUHAMMAD ASIR HOSSAIN CHOWDHURY

AI/ML PRACTITIONER | MACHINE LEARNING | DEEP LEARNING | PYTORCH

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LinkedIn | GitHub | Portfolio | Codeforces

PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate with hands-on projects in classical machine learning, PyTorch-based deep learning, NLP, and retrieval-augmented generation (RAG). I maintain public ML and PyTorch notebooks, have helped juniors and batchmates with programming and ML topics, and am preparing a machine learning session at East Delta University. Completed the DeepLearning.AI Machine Learning Specialization and maintain structured ML and PyTorch notes and notebooks on GitHub. Interested in explaining technical concepts through clear, step-by-step examples and practical code.

TECHNICAL SKILLS

Machine Learning: Regression, classification, Decision Trees, KNN, Naive Bayes, SVM, Random Forest, AdaBoost, XGBoost, K-Means, Hierarchical Clustering, DBSCAN, EDA, feature engineering, preprocessing, cross-validation, model evaluation.
Deep Learning: PyTorch, tensors and autograd, ANN/MLP, forward/backpropagation, loss functions, Momentum, AdaGrad, RMSProp, Adam, regularization, weight initialization, batch normalization, Dataset/DataLoader.
Computer Vision / NLP / GenAI: CNN fundamentals, padding/stride/pooling, RNN/LSTM fundamentals, text preprocessing, Word2Vec, RAG, embeddings, similarity search, LLM API integration.
Programming / Tools: Python, C/C++, Go, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, Streamlit, Gemini API, Groq/Llama 3, Git/GitHub.

SELECTED AI/ML PROJECTS

- **Heart Disease Classification | Scikit-learn** — Built a leakage-conscious preprocessing pipeline with ColumnTransformer, imputation, scaling and one-hot encoding; tuned Random Forest with RandomizedSearchCV and compared it with Logistic Regression. Test results: 87.50% accuracy, 92.16% disease-class recall, 0.9274 ROC-AUC. Educational project, not a diagnostic tool.
- **Fashion-MNIST Classification | PyTorch** — Implemented a fully connected neural network for 10-class clothing-image classification, using normalized tensors, CrossEntropyLoss, Adam, backpropagation and a separate test set; achieved 89.19% test accuracy.
- **PDF Question Answering System | RAG** — Built a Streamlit PDF Q&A app with document parsing, chunking, SentenceTransformers embeddings, NumPy similarity search and Groq/Llama 3 response generation.
- **AI Note Summarizer & Quiz Generator** — Developed a Streamlit and Gemini API app that turns uploaded note images into summaries and difficulty-based quizzes, evaluates answers, and provides audio summaries.

LEARNING RESOURCES & KNOWLEDGE SHARING

Maintain structured, topic-based notebooks and notes for ML (regression, classification, preprocessing, clustering and PCA) and PyTorch (perceptrons, autograd, backpropagation, optimizers, normalization and data loading). ML materials | PyTorch materials

EDUCATION & CERTIFICATION

B.Sc. in Computer Science & Engineering | East Delta University, Chattogram | Sep 2023 - Sep 2027 (expected)

Higher Secondary Certificate in Science | Government City College, Chattogram | 2020–22 | Grade: 5.00 out of 5.00

Secondary School Certificate in Science | Chattogram Collegiate School, Chattogram | 2018–20 | Grade: 5.00 out of 5.00

Machine Learning Specialization | DeepLearning.AI / Stanford University

LEADERSHIP & COMMUNITY EXPERIENCE

Campus Ambassador, Programming Hero (Feb 2026 - Present) - Represent the platform in EDU student community and promote course engagement.

Sub-Executive, EDU Computer Club (May 2026 - Present) | Sub-Executive, EDU Robotics Society (Nov 2025 - Present) - Support student community initiatives, events and society activities.

AWARDS & ADDITIONAL INFORMATION

Winner, 3 Minute Thesis - EDU SoSET Research Day

Honorable Mention, ICPC Online Preliminary 2025

6th Place, EDU Intra-University Programming Contest 2024

Hult Prize 2026 Finalist (Team INNOVxAMOR)

EXTRACURRICULAR ACTIVITIES

Cadet, Bangladesh National Cadet Corps (BNCC), Chattogram Collegiate School | 01/2018 – 09/2019
• Participated in drills, training, and community service activities, developing discipline, teamwork, and leadership skills.

Participant, 9th Dutch-Bangla Bank Prothom Alo Bangladesh Physics Olympiad | 2019 | Chattogram, Bangladesh
•Represented school at the national-level physics competition, applying strong problem-solving and logical reasoning skills.